

DAY 4 – SQL CONSTRAINTS

USE slatask;

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY,  
    product_name VARCHAR(50),  
    price INT  
);
```

```
INSERT INTO orders VALUES (1,'Laptop',50000);
```

```
INSERT INTO orders VALUES (2,'Mouse',500);
```

```
INSERT INTO orders VALUES (3,'Keyboard',1500);
```

```
INSERT INTO orders VALUES (4,'Laptop',52000);
```

```
INSERT INTO orders VALUES (5,'Mouse',700);
```

-- COUNT

```
SELECT COUNT(*) AS total_orders FROM orders;
```

OUTPUT

	total_orders
▶	5

-- sum

```
SELECT SUM(price) AS total_price FROM orders;
```

OUTPUT

	total_price
▶	104700

-- AVG

```
SELECT AVG(price) AS avg_price FROM orders;
```

OUTPUT

	avg_price
▶	20940.0000

-- Group by

```
SELECT product_name, SUM(price) AS total_sales
FROM orders
GROUP BY product_name;
```

OUTPUT

	product_name	total_sales
▶	Laptop	102000
	Mouse	1200
	Keyboard	1500

```
-- MAX and min
```

```
SELECT MAX(price) AS max_price, MIN(price) AS min_price
FROM orders;
```

OUTPUT

	max_price	min_price
▶	52000	500

Sort grouped data by total sales descending

```
SELECT product_name, SUM(price) AS total_sales
FROM orders
GROUP BY product_name
ORDER BY total_sales DESC;
```

OUTPUT

	product_name	total_sales
▶	Laptop	102000
	Keyboard	1500
	Mouse	1200

```
-- Create database
```

```
CREATE DATABASE companyDB;
```

```
USE companyDB;
```

```
-- create employee table with constraints
```

```
CREATE TABLE employees (
    id INT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    salary INT DEFAULT 30000,
```

email VARCHAR(50) UNIQUE

);

-- insert 5 records

INSERT INTO employees VALUES (1,'Arun',40000,'arun@gmail.com');

INSERT INTO employees VALUES (2,'Priya',35000,'priya@gmail.com');

INSERT INTO employees VALUES (3,'Karthik',30000,'karthik@gmail.com');

INSERT INTO employees VALUES (4,'Divya',45000,'divya@gmail.com');

INSERT INTO employees VALUES (5,'Rahul',30000,'rahul@gmail.com');

OUTPUT

	id	name	salary	email
▶	1	Arun	40000	arun@gmail.com
	2	Priya	35000	priya@gmail.com
	3	Karthik	30000	karthik@gmail.com
	4	Divya	45000	divya@gmail.com
	5	Rahul	30000	rahul@gmail.com
*	NULL	NULL	NULL	NULL

-- try duplicaate email

INSERT INTO employees VALUES (6,'Sam',38000,'arun@gmail.com');

OUTPUT

✖	66	16:29:46	INSERT INTO employees VALUES (6,'Sam',38000,'arun@gmail.com')	Error Code: 1062. Duplicate entry 'arun@gmail.com' for key 'employees....	0.031 sec
---	----	----------	---	---	-----------

-- Drop table

DROP TABLE employees;

✔	67	16:30:48	DROP TABLE employees	0 row(s) affected
---	----	----------	----------------------	-------------------

-- recreate table with check constraint

CREATE TABLE employees (

id INT PRIMARY KEY,

name VARCHAR(50) NOT NULL,

salary INT CHECK (salary > 0),

email VARCHAR(50) UNIQUE

);

-- create product table

```
CREATE TABLE products (  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    price INT  
);
```

-- Add new column stock

```
ALTER TABLE products  
ADD stock INT;
```

OUTPUT

	id	name	price	stock
*	NULL	NULL	NULL	NULL

-- Rename column price → unit_price

```
ALTER TABLE products  
RENAME COLUMN price TO unit_price;
```

-- Insert 5 products

```
INSERT INTO products VALUES (1,'Laptop',50000,10);  
INSERT INTO products VALUES (2,'Mouse',500,50);  
INSERT INTO products VALUES (3,'Keyboard',1500,30);  
INSERT INTO products VALUES (4,'Monitor',12000,15);  
INSERT INTO products VALUES (5,'Printer',8000,8);
```

OUTPUT

	id	name	unit_price	stock
▶	1	Laptop	50000	10
	2	Mouse	500	50
	3	Keyboard	1500	30
	4	Monitor	12000	15
	5	Printer	8000	8
*	NULL	NULL	NULL	NULL

-- Update stock of one product

```
UPDATE products
```

```
SET stock = 25
```

```
WHERE id = 3;
```

OUTPUT

	3	Keyboard	1500	25
--	---	----------	------	----

-- Delete one product

```
DELETE FROM products
```

```
WHERE id = 5;
```

OUTPUT

	id	name	unit_price	stock
▶	1	Laptop	50000	10
	2	Mouse	500	50
	3	Keyboard	1500	25
	4	Monitor	12000	15
★	NULL	NULL	NULL	NULL